

# Blockchain Technology

Mid Term Examination Solutions (E1PY214T)

## Section A: Network Models

### 1. Public vs. Private Blockchain [cite: 543]

| Feature         | Public Blockchain                     | Private Blockchain                            |
|-----------------|---------------------------------------|-----------------------------------------------|
| Access          | Anyone can join/read (Permissionless) | Restricted to authorized users (Permissioned) |
| Control         | Decentralized                         | Centralized by one organization               |
| Speed           | Slower due to large network           | Faster due to fewer nodes                     |
| Confidentiality | Low (Transparent)                     | High                                          |

### 2. Permissionless vs. Permissioned Models [cite: 543]

- **Permissionless:** Open to everyone, lacks a central authority, and typically uses resource-intensive consensus like Proof of Work (PoW). It is highly secure but less scalable.
- **Permissioned:** Users must be invited to join. It is often chosen for business applications because it offers better confidentiality, control over who can validate transactions, and higher scalability.

## Section B: Consensus Mechanisms

### 3. Delegated Proof of Stake (DPoS) [cite: 543]

DPoS is a consensus algorithm where stakeholders elect "delegates" or "witnesses" to validate blocks on their behalf. This voting system makes it one of the fastest consensus models, as only a small number of trusted nodes are responsible for maintaining the ledger, significantly reducing transaction times.

### 4. Consensus Algorithms and Agreement [cite: 543]

Consensus algorithms achieve agreement by ensuring all nodes in the network converge on a single version of the truth. They ensure security and fairness by requiring nodes to prove they have performed a specific task (PoW) or hold a stake (PoS) before they can suggest changes to the ledger.

## **Section C: Advanced Security**

### **5. Byzantine Fault Tolerance (BFT) [cite: 545]**

BFT handles malicious nodes by allowing the network to reach consensus even if some nodes fail or act dishonestly. As long as at least two-thirds (2/3) of the nodes are honest and functioning correctly, the system can bypass the "Byzantine" (malicious) nodes and ensure the reliability and security of the transactions.